

J/ψ radiative corrections: comparison of the two calculations

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“NLO paper” is *Fully accounted NLO electromagnetic radiative corrections to the width of J/ψ decay*; “the note” is *Radiative Corrections to the J/ψ photoproduction reaction*; Ref. [10] of the NLO paper is Ehlotzky & Mitter, *Nuovo Cimento* **55A** (1968) 181, abbreviated E&M.

1 The closed-form total correction is missing from the NLO paper

The NLO paper gives δ_{self} (8), δ_{vert} (12), δ_{soft} (13), δ_{hard} only as an unevaluated integral (14), and the exact Bardin–Shumeiko result (28). There is no compact closed form for the total ultrarelativistic correction — the quantity an experiment applies. It is E&M Eq. (26) with the missing term restored:

$$\delta(m_V, \omega_{cut}) = -\frac{4\alpha}{\pi} \left[\left(\frac{1}{2} - \ln \frac{m_V}{m_\ell} \right) \left(\ln r + \frac{1}{3} \right) + \frac{1}{2} \left(L_2(r) - \frac{\pi^2}{6} \right) + \frac{17}{72} + R(r) + \underbrace{\frac{r^2}{16} - \frac{3}{8}r}_{\text{missing from [10]}} \right]$$

$$r = \frac{2\omega_{cut}}{m_V}, \quad R(r) = \left(r - \frac{r^2}{4} - \frac{3}{4} \right) \left[\ln \left(\frac{m_V}{m_\ell} \sqrt{1-r} \right) - \frac{1}{2} \right], \quad L_2(x) = - \int_0^x \frac{\ln(1-t)}{t} dt$$

For muons add the electron-loop vacuum polarization, $= \delta_{self}$ of Eq. (8) with m_e . The missing piece is equivalently $+(\alpha/\pi)(3r/2 - r^2/4)$; it vanishes as $r \rightarrow 0$ and reaches $5\alpha/4\pi = 0.0029$ at $r = 1$.

Status of this formula. It is not exact in the lepton mass, and not the $m_\ell = 0$ limit either: the mass is kept in the logarithms while terms of relative order $(m_\ell/E)^2$, $E = m_V/2$, are dropped. E&M state this above their Eq. (22). Since $(m_e/E)^2 = 1.1 \times 10^{-7}$ and $(m_\mu/E)^2 = 4.7 \times 10^{-3}$, at $r = 1$ it is off by 1×10^{-7} for electrons and -7×10^{-5} for muons against the exact Eq. (28).

Where it should go: **Section IV**, immediately after Eq. (14) — the closed form *is* the evaluation of that integral combined with Eqs. (8), (12) and (13). Section IV opens by promising “compact expressions for all contributions to decay width”; compact expressions follow for every individual contribution but not for the sum, which is the only physical combination. Eq. (16), $\delta_{FSIC} = 3\alpha/4\pi$, then becomes a corollary — the $r \rightarrow 1$ limit less δ_{self} — and Figure 4 acquires the function it plots. Since Section IV also states that the URA is “applicable for $\ell = e$ ”, the electron expression should be the working formula, with the muon one alongside, its $(m_\mu/E)^2$ error flagged and Eq. (28) named as the fallback.

2 The missing term originates in Ref. [10]

E&M Eq. (23), the differential spectrum, is correct and agrees with Eq. (14) of the NLO paper. The term was lost when that spectrum was integrated analytically in 1968, so Eq. (24) and Eq. (26) of

[10] are both incomplete. The note transcribed those equations faithfully and inherited the omission; it has now been corrected (Section 6).

Where to record this: in Section IV, immediately after the closed-form expressions proposed above. Those expressions *are* E&M Eq. (26) with the term restored, so one paragraph placed there attributes the formula, records the correction, and explains the numerical comparison with [10] all at once. Suggested wording:

The expressions above are Eq. (26) of Ref. [10] with the term $(\alpha/\pi)(3r_{max}/2 - r_{max}^2/4)$ restored. The differential spectrum, Eq. (23) of [10], is correct and agrees with our Eq. (14); the term was lost in its analytic integration, so that Eq. (24), Eq. (26) and Table I of [10] are all low by that amount — $5\alpha/4\pi = 0.0029$ at $r_{max} = 1$, and below the precision quoted in [10] for $r_{max} \lesssim 0.1$.

3 Table I of Ref. [10] was computed with the incomplete formula

This matters for the validation in Section IV, because Table I of [10] is not a correct benchmark. It was computed from E&M’s own Eq. (26), and therefore carries the same omission.

The evidence is not one line but the whole table. Taking the δ column at $r_{max} = 1.00$, all fourteen mass rows, both channels — 28 entries:

agreement with the δ printed in Table I of [10]	electrons	muons
E&M Eq. (26) <i>as published</i> (term absent)	12/14 rows exact	12/14 rows exact
E&M Eq. (26) <i>corrected</i>	0/14	0/14

The two misses in each channel are last-digit rounding, not disagreement; the corrected formula misses every row by $\simeq 0.003$. So Table I of [10] is low by $(\alpha/\pi)(3r_{max}/2 - r_{max}^2/4)$ throughout. The agreement reported in the NLO draft at $\omega/m_e = 0.01$ holds only in the small- r_{cut} rows, where the missing term falls below the three decimals quoted in [10].

Two ways to handle this:

(1) State it in the NLO paper. The paragraph suggested in Section 2 above already covers Table I of [10]. The sentence claiming the comparison “succeeds” should then become a pointer to it, since as written it asserts agreement with a table that is itself in error.

(2) Recompute Table I at $m_V = M(J/\psi)$. Since [10] can no longer serve as the validation target, a table at the physical mass is more useful: it demonstrates the ω -independence just as well, and the numbers are ones a J/ψ experiment can use directly. For $m_V = 3.0969$ GeV:

r_{cut}	$\lg(\omega/m_\ell)$	e^-e^+			$\mu^-\mu^+$		
		δ_R	δ_H	δ	δ_R	δ_H	δ
0.10	−6	−1.577194	1.481750	−0.095444	−0.382419	0.376596	−0.005823
	−4	−1.225925	1.130481	−0.095444	−0.259274	0.253452	−0.005823
	−2	−0.874655	0.779212	−0.095443	−0.136130	0.130328	−0.005801
0.25	−6	−1.577194	1.540558	−0.036636	−0.382419	0.396794	+0.014375
	−4	−1.225925	1.189288	−0.036636	−0.259274	0.273649	+0.014375
	−2	−0.874655	0.838019	−0.036636	−0.136130	0.150526	+0.014396
0.50	−6	−1.577194	1.576919	−0.000275	−0.382419	0.408881	+0.026462
	−4	−1.225925	1.225649	−0.000275	−0.259274	0.285736	+0.026462
	−2	−0.874655	0.874380	−0.000275	−0.136130	0.162613	+0.026483
1.00	−6	−1.577194	1.603329	+0.026135	−0.382419	0.416436	+0.034017
	−4	−1.225925	1.252060	+0.026135	−0.259274	0.293291	+0.034017
	−2	−0.874655	0.900791	+0.026136	−0.136130	0.170168	+0.034038

δ_R for the muon channel includes both self-energy loops. The $r_{cut} = 1.00$ electron entry, 0.026135, equals $\delta_{self} + 3\alpha/4\pi$ and matches the Bardin–Shumeiko value of Table III. The two approaches are not exclusive; (1) is needed in any case.

4 Figure 4 of the NLO paper and Figure 8 of the note

These are now the same picture. Figure 8 of the note has been redrawn against $r_{max} = 2k_{max}/M_V$ over the full physical range of each channel, so it carries the same content as Figure 4, which uses $\omega_{cut}/\omega_{max}$. The two abscissae are related by $r_{max} = (\omega_{cut}/\omega_{max})\beta^2$ with $\beta^2 = 1 - 4m_\ell^2/M_V^2$, and the curves agree to machine precision at every point sampled, in both channels.

5 Formula comparison: the points that carried a finding

note	NLO	
(7)	(6)	identical (verified symbolically)
(11),(12)	(8)+(12)+(13)	identical to machine precision
(13)	(14) integrand	identical
(14),(15),(19),(20)	—	were missing the term; corrected in the note
(20)	—	leading logarithm had m_e where m_μ belongs; corrected
(6)	(1)+(2)	$4m_\ell^2/m_V$ was not dimensionless; corrected
(16)	(15)	both take $k_{max}^{max} = M_V/2$ in the text; the exact end-point is $M_V/2 - 2m_\ell^2/M_V$. Corrected in the note; see below
—	(4)	polarization sum written as for a massless vector
—	(16)	FSIC $3\alpha/4\pi$; reproduced by the corrected closed form

On the last-but-two line: the NLO paper says the cut takes its maximum “at (15) or $r_{cut} = r_{max} = 1$ ”, and Table III is captioned ($r_{max} = 1$), but Eq. (15) itself gives $r_{cut}^{max} = 1 - 4m_\ell^2/m_V^2 = \beta^2$, which

is 0.99535 for muons. Taken literally, $\omega_{cut} = m_V/2$ makes the second radicand of ρ_1 in Eq. (29) negative and ρ_1 imaginary, in both channels. The labels should read $\omega_{cut} = \omega_{max}$, or $r_{cut} = \beta^2$.

6 Status of the note

Corrected: the missing term in Eqs. (14), (15), (19), (20); the leading logarithm of Eq. (20); the phase-space factor of Eq. (6); the maximum photon energy, which now carries the exact endpoint and the bound $r_{max} \leq \beta^2$ in Eq. (16); Fig. 7, recomputed at the ε stated in its caption; Fig. 8, redrawn against r_{max} over the full range. A short paragraph giving the correspondence between the variables of the two papers ($\varepsilon \leftrightarrow \omega$, $k_{max} \leftrightarrow \omega_{cut}$, $r_{max} \leftrightarrow r_{cut}$, $\ln(M_V/m_\ell) = L/2$) has been added after Eq. (17). Equation numbering is unchanged throughout.

Not yet examined: Eqs. (2)–(5), (17), (18) of the note, and the Bethe–Heitler formulas behind the –20% comparison of its Section VII.